

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show IP route ISPF

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _129_

Does R2 show up as an OSPF neighbor on R1? Yes _____

Does R2 show up as an OSPF neighbor on R3? No _____

What does this information tell you?

_This shows the interface S0/0/1 on R2 is still configured as a passive interface and there is no OSPF Hello packets are sent. R2 and R3 cannot form OSPF neighbor adjacency and R3 need to learn the the rout of of R2 through R1

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_R2(config) # router ospf 1

R2 (config-router)# no passive-interface s0/0/1

Re-issue the show ip route command on R3

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Accumulated=65, and it is equal to OSPF cost of the outgoing interface on R39 (serial0/0/1) and the destination outgoing interface of R2 (GigabitEthernet0/0, cost =1)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

By OSPF cost formula is a the interface with a speed 100 Mps or above will count as 1. If we increase the bandwidth , it allows OSPF routing process to calculate metrics and diffrenciater between higher speed link accurately

Step 2: Change the bandwidth for an interface.

- g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The accumulated OSPF cost is calculated by summing the costs of all outgoing interfaces along the routing path to the destination network. However, the cost for each individual interface could indicate the bandwidth capacity, determined by the formula: $100,000,000 / \text{bandwidth_in_bps}$. Moreover, the significant findings here may suggest that a default serial interface (1.544 Mbps) has a cost of 64, while a GigabitEthernet interface has a cost of 1. Furthermore, the key evidence may indicate that these cost values could demonstrate meaningful differences in path selection across the network topology. Cost drives path choice.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost may suggest that the resulting value of 781 reflects direct calculation from the interface configuration. Moreover, the bandwidth 128 command could indicate that the interface bandwidth is manually set to 128 kbps (128,000 bps). In light of this configuration, the OSPF cost formula ($100,000,000 / 128,000$) might demonstrate that the resulting cost for that single outgoing serial interface becomes 781. Furthermore, the significant evidence could suggest that this manual setting directly affects the accumulated cost metric.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The route path changed because the OSPF cost of the direct serial link between R1 and R3 was manually increased using the `ip ospf cost` command. Thus, the significant results may suggest that the direct path's accumulated cost metric is now higher than the indirect path traversing through R2. Given that the evidence demonstrates that the OSPF algorithm always prefers the route with the lowest accumulated cost, the results might indicate that it dynamically recalculates and chooses the path through R2. Additionally, the findings could suggest that this dynamic recalculation may demonstrate the algorithm's responsiveness to manually configured cost changes. Algorithm recalculates, selects lowest cost path.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling the router ID is important because it uniquely identifies the router within the OSPF domain. However, the significant evidence may suggest that a strategically assigned router ID could demonstrate much easier network management and troubleshooting of routing issues. In light of these findings, the key results might indicate that router ID assignments could influence which router becomes the Designated Router (DR) or Backup Designated Router (BDR) when interface priorities are equal. Notwithstanding this consideration, the evidence may suggest that multi-access networks could demonstrate particularly important reliance on router ID for the DR and BDR election process. Router ID determines DR and BDR roles.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election process is not a concern in this lab because the routers are connected using point-to-point serial links. Therefore, the important findings may suggest that DR and BDR elections could demonstrate relevance only on multi-access networks like Ethernet. Moreover, the evidence might indicate that on a point-to-point link, only two devices exist, suggesting that a DR/BDR appears unnecessary. Furthermore, the significant results may suggest that this architectural consideration could demonstrate that point-to-point configurations avoid unnecessary election overhead entirely. Point-to-point links need no DR or BDR.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive prevents OSPF routing updates, specifically Hello packets, from being sent out of that interface, while still allowing the network associated with that interface to be advertised to other OSPF routers. However, the key evidence may suggest that this configuration could demonstrate particular value for LAN interfaces connected to end devices like PCs. Given that the findings demonstrate that passive interfaces save network bandwidth and reduce processing load, the results might indicate that this approach could improve security by preventing unauthorized devices from forming OSPF adjacencies. Thus, the significant evidence may suggest that passive interface configuration could demonstrate a critical balance between advertisement capability and Hello packet suppression. Passive settings protect LAN segments effectively.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of Areas, such as Area 0 the backbone area, is the aspect of OSPF that makes it hierarchical. Moreover, the findings may suggest that dividing a large OSPF domain into multiple smaller areas could demonstrate that OSPF effectively restricts the flooding of link-state updates.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

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updates. _____

